

or 29.8 bps of the notional, the same price that we obtained for the 100 out payer swaption. On a notional of \$25mm, the premium is $\$25\text{mm} \times 29.8 \text{ bps} = \$74,500$.

The price of a 1 into 3 swaption straddle struck at-the-money forward (i.e., a strike of 5.644%) is

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A = 0.02310,$$

or 231.0 bps of the notional. On a notional of \$25mm, the premium is $\$25\text{mm} \times 231.0 \text{ bps} = \$577,500$. $\square$

5.4.4 Relationship Between Normal Implied Vol and Lognormal Implied Vol for At-the-Money Swaptions

There is a relationship between normal implied vol, σ_N , in the normal model and lognormal implied vol in Black's model. In this section, we examine that relationship for the case of swaptions that are struck at-the-money forward. For a given price of a swaption computed using the normal model with normal implied vol σ_N , let us now denote by σ_{LN} the lognormal implied vol that must be used in Black's model to result in the same swaption price. As is also shown in Appendix E, for at-the-money swaptions the relationship between normal implied vol and lognormal implied vol is

$$\sigma_N \approx F \times \sigma_{LN}, \quad (5.27)$$

where F denotes the underlying forward swap rate.³⁴ This approximation is widely used throughout the industry, although it is sometimes mistakenly believed to hold with equality. It is very important to note that the relationship in Equation 5.27 is an approximation. The relationship is very important actually because under the assumption that the normal model is correct (i.e., that normal implied vol σ_N is constant regardless of the level of rates), it dictates restrictions on the behavior of lognormal implied vol, σ_{LN} . In particular, it indicates that if normal implied vol is constant, then lognormal implied vol must go down when rates go up, and vice versa.

³⁴Equation 5.27 provides additional insight as to why we have harped on about the fact that σ_N is a measure of absolute volatility, whereas σ_{LN} is a measure of relative volatility. In accordance with Equation 5.27, for at-the-money forward swaptions if we take the relative volatility σ_{LN} and multiply by the level of the rate, F , we get (approximately) the level of absolute volatility, σ_N .

Example

Suppose that a certain forward swap rate is at 5% and further suppose that normal implied vol is 100 bps. If this forward rate rises to, say, 8%, then normal implied vol would remain at 100 bps under a normal model. But what does this imply about the behavior, then, of lognormal vol? If the forward swap rate is 5% and we have normalized implied vol of 100 bps, then according to Equation 5.27 lognormal implied vol is approximately 20%. If rates go up to 8% and normal implied vol remains at 100 bps, Equation 5.27 suggests that lognormal implied vol will have to decrease to around 12.5%. $\square$

Empirical Evidence

The approximation in Equation 5.27 and its implications on the behavior of lognormal implied vol under the assumptions of the normal model (i.e., that σ_N remains constant as rates move) is in fact characteristic of what we often observe in the rates markets.

Take a look now at Figure 5.13. The top half of this figure shows the relationship between lognormal implied vols and the level of forward swap rates. As shown, there is a significant relationship, and lognormal implied vol falls as rates go higher, and vice versa. In contrast, the bottom half of this figure shows the relationship between normal implied vol and the level of forward swap rates. Here we find that there is much less of a relationship (given the lower R^2) between the level of normal implied vol and the level of forward swap rates.

So what does this tell us? First, the low R^2 of the bottom panel is consistent with the assumption that normal implied vol is constant and does not vary with rates, as assumed by the normal model. Second, the top panel, which shows that there is a more significant relationship between lognormal implied vol and the level of rates, tends to refute the lognormal model. The lognormal model assumes that lognormal implied vol σ_{LN} should remain constant when rates move. Third, since the top panel is more consistent with the notion that lognormal implied vol varies inversely with the level of rates, this lends support for Equation 5.27, which is again consistent with the normal model. A direct implication of the assumption of constant basis point volatility in the normal model is that percentage volatility must be high in a low-rate environment and low in a high-rate environment.

Given such evidence in support of the normal model, it is not surprising, then, that it has become quite common in the market to speak of vols in terms of normal implied vol. In Table 5.1, we introduced the concept of a vol grid or swaption matrix of lognormal implied vols. It is now at least as common in the market to refer to a matrix of normal implied vols as seen in Table 5.3.

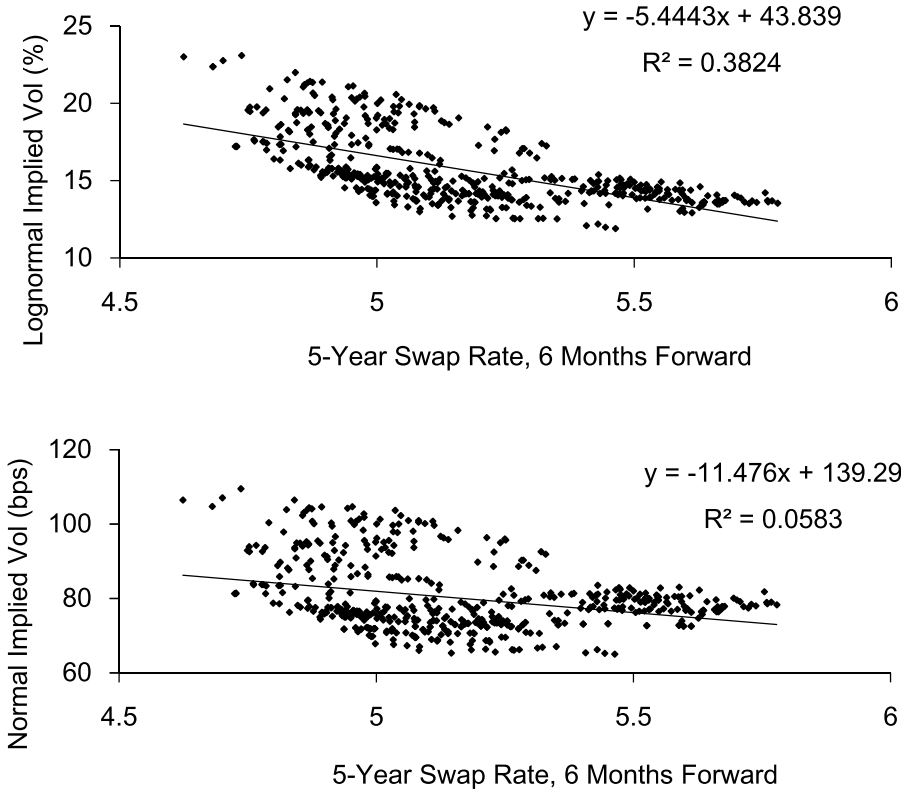


Figure 5.13: Regressing Lognormal Implied Vol on the Level of Rates (*top half*) Shows a Significant Relationship, Refuting the Black-Scholes Assumptions. Regressing Normal Implied Vol on the Level of Rates (*bottom half*) Shows a Less Significant Relationship, in Support of the Normal Model

Again, Black's model assumes constant percentage volatility for an underlying swap rate, and the numbers in the lognormal swaption matrix are given as percentages. By contrast, in the normal model, the basis point volatility of an underlying swap rate is assumed to be a constant, independent of the level of rates, and the numbers in the normal swaption grid represent absolute numbers of basis points.

A Better Approximation

Once again, it is important to note that the relationship in Equation 5.27 is an approximation. Moreover, it is a relationship that holds only for at-

		Underlying Swap (Tail)								
		1	2	3	5	7	10	15	20	30
E x p i r y	1m	94.1	114.0	119.1	122.7	118.0	110.6	99.8	92.8	85.5
	3m	99.7	119.7	123.7	125.9	121.0	113.8	102.3	95.0	88.9
	6m	111.9	123.4	124.6	125.8	120.9	113.7	102.6	95.3	89.9
	1y	121.0	124.2	124.0	123.5	119.4	112.9	101.3	94.3	89.5
	2y	125.7	124.2	122.5	119.5	115.0	109.0	97.5	91.0	85.7
	3y	122.2	120.5	118.2	114.9	111.1	105.2	94.2	87.4	81.6
	4y	118.2	116.4	114.0	110.0	106.4	100.8	89.4	82.3	77.8
	5y	115.0	113.2	110.2	105.8	101.9	96.9	85.2	78.7	73.5
	7y	106.2	103.7	101.0	96.5	93.3	88.4	77.9	71.5	67.1
10y	92.4	90.1	88.0	83.5	80.6	75.8	66.8	61.5	58.1	

Table 5.3: Normal Implied Swaption Vol Matrix

the-money forward straddles. A better (albeit infrequently used) approximation for the relationship between σ_N and σ_{LN} for at-the-money straddles is given by

$$\sigma_N \approx F \times \sigma_{LN} \times \left(1 - \frac{\sigma_{LN}^2 T}{24}\right). \quad (5.28)$$

Equation 5.28 provides a meaningfully better approximation of normal implied vol for cases in which the time to expiration T of the swaption is large and/or lognormal implied vol σ_{LN} is large. Once again, the approximation formulas in Equations 5.27 and 5.28 apply only to swaptions that are struck at-the-money forward.

Example

In Section 5.1.6, we priced up 1 into 3 payer and 1 into 3 receiver swaptions struck at-the-money forward in the context of our stylized example. Using a lognormal implied vol $\sigma_{LN} = 20\%$, we found that the value of each option was around 115.5 bps, and thus the at-the-money forward straddle is around 231 bps. In order to compute these prices, we needed to make use of the 1x4 swap rate, which was found to be 5.644%. We also found that the value of the annuity factor, A , was 2.5687. What is σ_N , normal implied vol, in the normal model?

The price of a 1 into 3 swaption straddle struck at-the-money forward in the normal model must be the same price as in the lognormal model. Thus, according to Equation 5.26, we must have

$$0.0231 = \sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A. \quad (5.29)$$

Substituting $A = 2.5687$ and $T = 1$ gives $\sigma_N = 0.01127$, or 112.7 bps. Alternatively, we can use Equation 5.27 to approximate σ_N as follows:

$$\sigma_N \approx F \times \sigma_{LN} = 0.05644 \times 0.20 = 112.9 \text{ bps},$$

close to the true answer of 112.7 bps. Alternatively, we can use Equation 5.28 to approximate σ_N more closely as follows:

$$\begin{aligned} \sigma_N &\approx F \times \sigma_{LN} \times \left(1 - \frac{\sigma_{LN}^2 T}{24}\right) \\ &= 0.05644 \times 0.20 \times \left(1 - \frac{0.20^2 \times 1}{24}\right) \\ &= 112.7 \text{ bps}, \end{aligned}$$

which is very close to the true answer obtained from Equation 5.29. $\square$

Table 5.4 provides a list of at-the-money forward swaption straddles, their premia and strikes, and the implied normal and lognormal vols. These premia and forward strike rates are based on the close of business on April 9, 2008. The figure also shows the approximation of the normal implied vol, using both methods in Equation 5.27 and Equation 5.28, showing the latter approximation formula results in smaller errors.

Basis Points per Day Vol

Another concept related to normalized vol that is frequently employed is known as *basis points per day vol*. Basis points per day vol is given by

$$\frac{\sigma_N}{\sqrt{252}} \approx \frac{F \times \sigma_{LN}}{\sqrt{252}}. \quad (5.30)$$

The number 252 represents the number of trading days in a year. Throughout the industry, basis points per day vol is often regarded as the number of basis points per day that the underlying swap rate must move on average in order for the buyer of the swaption (who pays an upfront premium) to break even by delta hedging the option daily. This interpretation is in general not true; only under the most special of circumstances is this interpretation

Straddle	Price (bps)	ATMF Strike (F)	σ_N	σ_{LN}	$F \times \sigma_{LN}$	Error	$F \times \sigma_{LN} \times (1 - \frac{\sigma_{LN}^2 T}{24})$	Error
3m1y	43.8	2.373%	111.7	47.2%	111.9	0.259	111.7	0.000
3m2y	103.4	2.614%	134.1	51.5%	134.5	0.360	134.1	0.011
3m5y	264.7	3.486%	145.0	41.7%	145.2	0.269	145.0	0.007
3m10y	429.0	4.181%	130.6	31.3%	130.7	0.145	130.6	0.012
6m1y	62.1	2.404%	112.4	47.0%	113.0	0.534	112.4	0.015
6m2y	138.2	2.739%	126.8	46.5%	127.4	0.587	126.9	0.012
6m5y	349.9	3.604%	135.6	37.8%	136.0	0.412	135.6	0.008
6m10y	572.9	4.259%	123.7	29.1%	123.9	0.208	123.6	0.011
1y1y	88.7	2.672%	116.2	43.9%	117.2	0.937	116.2	0.002
1y2y	184.3	3.079%	122.7	40.1%	123.5	0.827	122.7	0.001
1y5y	443.4	3.850%	125.2	32.7%	125.7	0.546	125.2	0.013
1y10y	735.5	4.421%	115.9	26.3%	116.2	0.318	115.9	0.017
2y1y	121.7	3.496%	116.5	33.6%	117.6	1.097	116.4	0.011
2y2y	240.2	3.787%	117.2	31.2%	118.1	0.959	117.2	0.001
2y5y	557.3	4.306%	115.9	27.1%	116.7	0.718	115.9	0.004
2y10y	933.8	4.718%	108.9	23.2%	109.4	0.506	108.9	0.015
5y1y	148.6	4.725%	102.3	21.9%	103.3	1.006	102.3	0.022
5y2y	289.0	4.816%	101.6	21.3%	102.5	0.954	101.6	0.014
5y5y	655.7	5.015%	99.4	20.0%	100.2	0.824	99.4	0.010
5y10y	1,102.1	5.193%	94.4	18.3%	95.0	0.678	94.4	0.015
10y1y	136.5	5.334%	85.6	16.2%	86.5	0.914	85.5	0.033
10y2y	265.0	5.372%	84.9	16.0%	85.8	0.928	84.9	0.015
10y5y	593.9	5.423%	82.6	15.4%	83.5	0.833	82.6	0.009
10y10y	1,002.1	5.355%	78.9	14.9%	79.6	0.732	78.9	0.002

Table 5.4: At-the-Money Forward Swaption Straddles and Their Implied Vols from both Black's Model and the Normal Model. Note: Rounding Employed in This Table May Make Certain Computations Appear Inaccurate

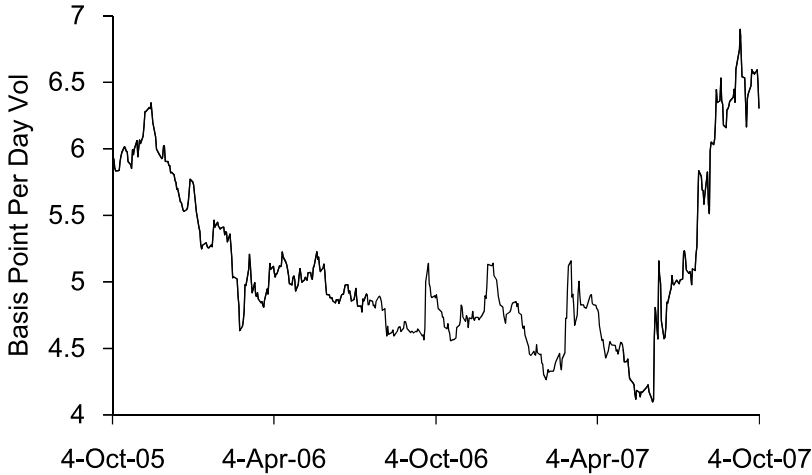


Figure 5.14: Basis Points per Day Vol for 6-Month into 5-Year Swaptions

valid.³⁵ Nonetheless, basis points per day vol remains a useful concept and one that is often used to gauge the relative richness or cheapness of a swaption. Figure 5.14 shows an example of a history of basis points per day vol for a swaption. As shown in this figure, after a prolonged period of declining implied volatilities (as a function of reduced structural demand for vol by certain market participants and a long period of low realized vol in the rates markets), vol increased dramatically during the credit crunch in the summer of 2007.

³⁵To see that this interpretation is not valid in general, let's consider two cases in which the buyer of the option delta hedges. In the first case, the underlying increases each day by an amount equal to the bps per day vol. In the second case, the underlying oscillates up and down one day after the next, always hovering around the strike. As indicated in Appendix E, under the normal model, the gamma of a payer or receiver is given by

$$A \times \frac{n(d)}{\sigma_N \sqrt{T}}$$

and thus is highest when the underlying is at the strike and diminishes as the underlying moves away from the strike. In the first case the gamma of the option (under a normal model) decreases steadily, whereas in the second it is relatively high, indicating the profits from delta hedging would be higher.

5.4.5 Explaining Skew: The Relationship Between Normal Implied Vol and Lognormal Implied Vol for Off-the-Money Swaptions

One of the troubling issues of the lognormal model that we identified earlier is the existence of skew that this model exhibits. That is, suppose we look at market prices on different swaptions on the same underlying swap rate and time to maturity, allowing only strikes to differ. What we often find is that the lognormal implied volatility is a decreasing function of strike, referred to as downward skew. Of course, if Black's model is correct, there should be flat skew, meaning that the lognormal implied vol should be the same regardless of the strike.

As we mentioned previously, one of the pleasing features of the normal model is that the assumption of constant normal vol actually dictates that lognormal implied vol must exhibit downward skew. We showed this earlier in Figure 5.8. In this section, we hope to provide some insight as to why this is the case.

Let's start by reviewing a few pricing results that we found in Section 5.4.3. In the example of this section, we looked at pricing up some 1 into 3 swaptions using the normal model. We considered our stylized example with the swap curve as in Table 5.2. The at-the-money strike is 5.644%. We assumed that σ_N was a constant 112.7 bps, regardless of the strike.

Using the normal model, we found that the price of a 100 out 1 into 3 receiver swaption (i.e., a 1 into 3 receiver swaption with a strike 100 bps below the forward) was 29.8 bps. In addition, we found that the price of a swaption straddle struck at-the-money forward was 231 bps. Thus, we can conclude that the price of a 1 in 3 receiver swaption struck at-the-money forward must be 115.5 bps.³⁶ We've collected these prices and used the normal model to price up a few additional strikes and have presented these results in Table 5.5.

So let's consider the following experiment. Suppose now we use Black's formula to price up the same swaptions as in Table 5.5. For each of the strikes listed in this table, we will set the value of the swaption equal to the price obtained using the normal model, then compute the lognormal implied vol that equates to that price. For example, if we assume that σ_N is 112.7 bps, then the price for a 1 into 3 receiver swaption struck at 4.644% is 29.8 bps. If we use Black's formula to price up the same swaption and want to come up with the same price of 29.8 bps, then we must set σ_{LN} to 22.0%. The results

³⁶From swaption parity, we know that the value of a receiver swaption struck at-the-money forward must be equal to the value of a payer swaption struck at-the-money forward. Thus, either the payer or receiver must be half of the at-the-money forward straddle price.

Strike	Price (bps)	σ_N
3.644%	4.4	112.7
4.644%	29.8	112.7
5.644%	115.5	112.7
6.644%	286.6	112.7
7.644%	518.1	112.7

Table 5.5: Prices of Receiver Swaptions of Various Strikes Under the Normal Model Using the Swap Curve in Table 5.2 and a Constant $\sigma_N = 112.7$ bps

Strike	Price (bps)	σ_N	σ_{LN}
3.644%	4.4	112.7	24.7%
4.644%	29.8	112.7	22.0%
5.644%	115.5	112.7	20.0%
6.644%	286.6	112.7	18.4%
7.644%	518.1	112.7	17.1%

Table 5.6: Solving for the Lognormal Implied Vol in Order to Equate to the Prices Obtained from the Normal Model in Table 5.5

of this experiment are shown in Table 5.6. Comparing results for the other strikes we see that as the strike goes down, the level of lognormal implied vol must go up, and vice versa. These results are displayed graphically in Figure 5.15.³⁷ That is, under the assumption of constant normal implied vol, we must have downward lognormal skew. Thus, the normal model is consistent with the notion of downward lognormal skew sometimes observed in the market, as was displayed in Figure 5.7. Of course, the lognormal model assumes that lognormal implied vol, which in this case is 20% for the at-the-money forward strike, does not vary with the level of the strike.

5.4.6 The Normal Model: The Industry Standard

The normal model for pricing interest rate options is by no means perfect. For example, it allows for the possibility of negative interest rates and does not explain various skew patterns that are sometimes observed in the market.

³⁷The manner in which we have generated the skew pattern in Figure 5.15 is identical to the procedure that was used to produce the skew pattern in Figure 5.8 earlier in this chapter.

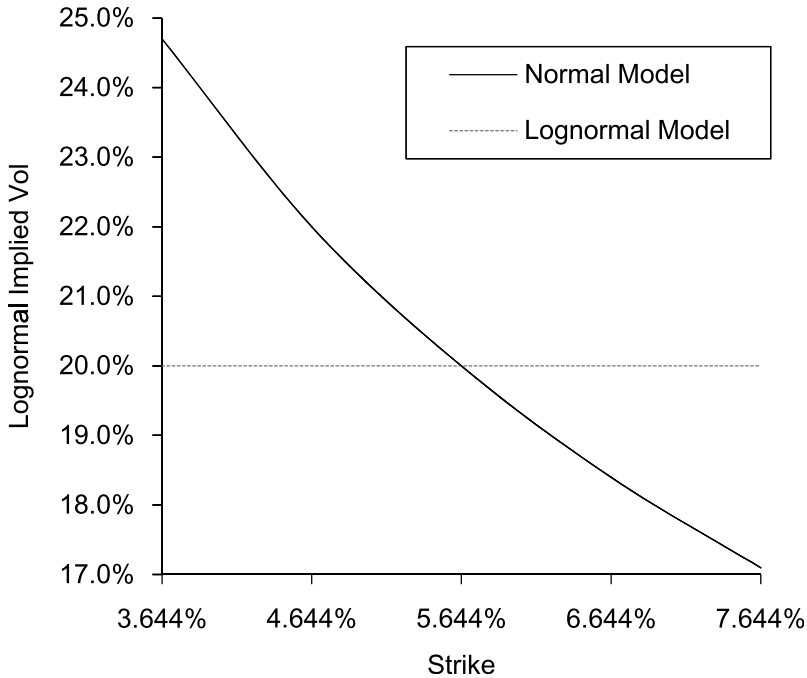


Figure 5.15: Flat Skew of the Black (Lognormal) Model and Downward Skew of the Normal Model

Moreover, there is substantial evidence that at least one of the assumptions of the normal model, namely, that normal implied vol is a constant, is violated. Nonetheless, the normal model appears to do a better job of depicting what is observed in the market and for the pricing of interest rate options than does Black's model.³⁸

³⁸Black's model kind of reminds me of the Jack Nicholson character Colonel Jessup in the movie *A Few Good Men*. You could imagine Black's model sitting up there on the stand, defending itself from naysayers who say its methods for pricing interest rate derivatives are inappropriate:

I have a greater responsibility than you could possibly fathom. . . . You have the luxury of not knowing what I know. . . . And my existence, while grotesque and incomprehensible to you, saves lives. Because deep down, in places you don't talk about at parties, you want me. . . . you need me. . . . We use words like honor, code, loyalty. We use these words as the backbone of a life spent defending something. You use them as a punch line. I have neither the time nor the inclination to explain myself to the man who rises and